

Comments in Java

Using Single-Line and Multi-Line Comments in Java

In Java, comments are used to add explanatory notes within the code. Comments are ignored by the Java compiler, and they help developers understand the code, add explanations, or temporarily disable parts of the code. Java supports two types of comments:

1. **Single-Line Comments**
 2. **Multi-Line Comments**
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1. Single-Line Comments

Single-line comments begin with two forward slashes (`//`). Everything following `//` on that line is considered a comment and will be ignored by the compiler.

- **Syntax:**

```
// This is a single-line comment
```

- **Usage:** Single-line comments are used for brief explanations or notes that fit on one line. They are ideal for commenting out small sections of code or adding quick notes.

Example:

```
public class Main {  
    public static void main(String[] args) {  
        // This is a single-line comment explaining the next line of code  
        int x = 10; // Initializing variable x with value 10  
  
        System.out.println("Value of x: " + x); // Print the value of x  
    }  
}
```

- **Explanation:**
 - The first `//` comment explains the code that follows it.
 - The second `//` comment is placed at the end of a statement and explains the purpose of that line.
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2. Multi-Line Comments

Multi-line comments (also called block comments) start with `/*` and end with `*/`. Everything between `/*` and `*/` is treated as a comment, even if it spans multiple lines.

- **Syntax:**

```
/* This is a multi-line comment  
   that spans more than one line */
```

- **Usage:** Multi-line comments are used when you need to comment out larger sections of code or write detailed explanations that span several lines.

Example:

```
public class Main {  
    public static void main(String[] args) {  
        /* This is a multi-line comment  
           It is used to explain the program in more detail.  
           We will initialize two variables and print their sum. */  
  
        int a = 5;  
        int b = 3;  
  
        // Printing the sum of a and b  
        System.out.println("Sum of a and b: " + (a + b));  
    }  
}
```

- **Explanation:**
 - The multi-line comment spans multiple lines and provides a detailed explanation of the program's logic.
 - The single-line comment at the end of the program explains what happens in the next line of code.

Best Practices for Using Comments in Java

- **Single-Line Comments:**
 - Use single-line comments to explain simple code or to add brief remarks.
 - Place comments above the code or at the end of a line of code.
- **Multi-Line Comments:**

- Use multi-line comments to explain more complex logic or to disable large blocks of code during debugging.
- Keep multi-line comments concise and meaningful.
- **Avoid Overusing Comments:**
 - Comments should explain the "why" behind the code, not the "what" (the code itself should be self-explanatory as much as possible).
 - Avoid commenting obvious code or trivial operations.